

ENTERPRISE IDS/IPS & SIEM PIPELINE

Integration Project

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Role	Junior SOC Analyst / Penetration Tester
Platform	VMware Workstation — 4-VM Lab Environment
Stack	pfSense · Suricata · Wazuh · rsyslog
Date	April 2026
Status	Completed — Full-Stack Detection Validated

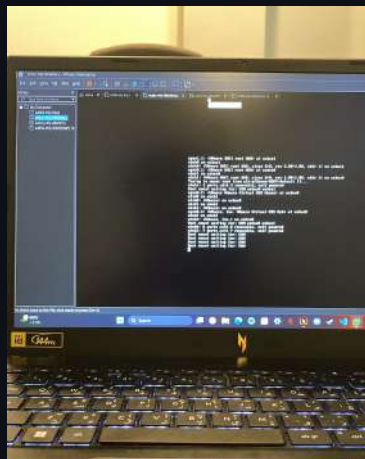


Fig 1 — VMware Workstation with 4-node lab: Kali, pfSense Firewall, Ubuntu (Suricata), Windows 10

1. Project Overview

This project documents the design and deployment of a production-grade, three-tier Security Operations Center (SOC) defense stack built entirely within a self-hosted VMware lab. The goal was to simulate a realistic enterprise threat detection environment — integrating a perimeter firewall, network-layer intrusion detection/prevention, centralized log management, and endpoint visibility — then validate the full pipeline by executing live attack simulations.

2. Lab Architecture

The lab consisted of four virtual machines interconnected on an isolated internal network:

VM Node	OS / Role	Key Function
AMIN-HQ-KALI	Kali Linux — Attacker	Payload delivery, Sliver C2, path traversal attacks
Amin-HQ-FIREWALL	pfSense — Perimeter Gateway	Network segmentation, syslog forwarding to Wazuh
Amin-HQ-UBUNTU	Ubuntu — IDS/IPS Host	Suricata IDS/IPS engine, rsyslog pipeline to Wazuh
AMIN-HQ-WINDOWS10	Windows 10 — Endpoint Target	Wazuh agent, EventLog Analyzer, C2 beacon target

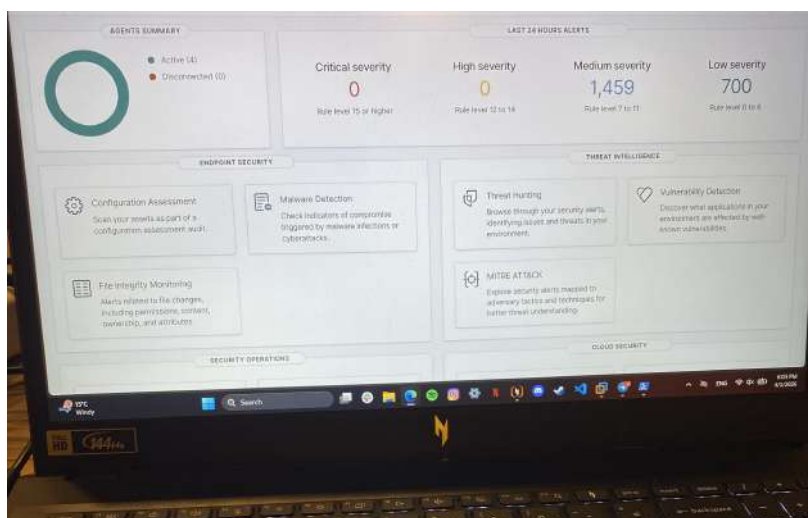


Fig 2 — Wazuh SIEM Overview Dashboard: 4 active agents, 1,459 medium-severity and 700 low-severity alerts logged in 24 hours. Modules active: FIM, Malware Detection, Threat Hunting, MITRE ATT&CK mapping, Vulnerability Detection.

3. Component Configuration

3.1 pfSense Perimeter Gateway & Syslog Forwarding

pfSense was deployed as the network's perimeter gateway, handling NAT, interface segmentation, and firewall rule management. A custom firewall ruleset was engineered to permit syslog traffic from pfSense to the Wazuh manager, enabling real-time perimeter event visibility within the SIEM. The filter reload output confirmed successful rule creation for syslog forwarding.

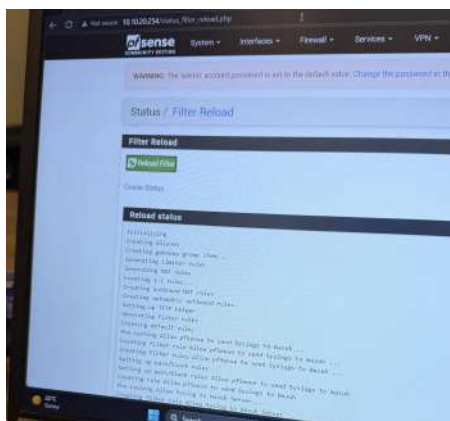


Fig 3 — pfSense Filter Reload output confirming creation of syslog forwarding rules to the Wazuh server.

3.2 Suricata IDS/IPS Engine

Suricata was deployed on the Ubuntu host in IDS/IPS/NSM/FW mode using the af-packet capture method. Key engineering challenges included resolving virtual network checksum validation issues (bypassed via configuration override) and diagnosing YAML/JSON serialization conflicts in the rsyslog pipeline. Once resolved, Suricata's eve.json log feed was consumed by the Wazuh agent and forwarded to the centralized SIEM.

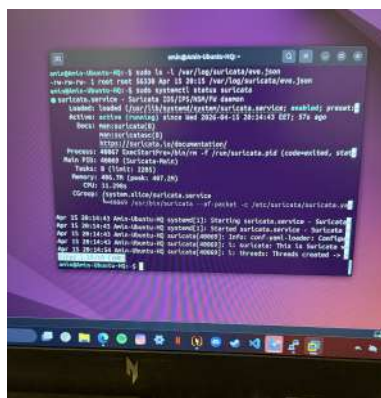


Fig 4 — Suricata service confirmed active (running) on Ubuntu-HQ, processing eve.json at /var/log/suricata/.

4.1 Suricata Network-Layer Alerts

Attacker	Victim-ID	Web Application Attack	10.10.20.131	33282	10.10.20.254	10	Summary
1	1	Web Application Attack	10.10.20.131	33282	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
2	2	Web Application Attack	10.10.20.131	33280	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
3	3	Web Application Attack	10.10.20.131	33286	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
4	4	Web Application Attack	10.10.20.131	33284	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
5	5	Web Application Attack	10.10.20.131	33282	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
6	6	Web Application Attack	10.10.20.131	33282	10.10.20.254	10	1.2024584 ET SCAN Possible Denial of Service Observed
7	7	Generic Protocol Command Decode	10.10.20.264	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
8	8	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
9	9	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
10	10	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
11	11	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
12	12	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
13	13	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
14	14	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
15	15	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
16	16	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
17	17	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
18	18	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed
19	19	Generic Protocol Command Decode	10.10.20.254	0	10.10.20.131	9	1.2024584 ET SCAN Possible Denial of Service Observed
20	20	Generic Protocol Command Decode	10.10.20.131	0	10.10.20.254	9	1.2024584 ET SCAN Possible Denial of Service Observed

4.2 Wazuh Threat Hunting — Kali Linux Detection

845 hits

Apr 15, 2023 @ 20:22:41:232 - Apr 15, 2028 @ 20:22:41

100 available fields | 1 fields sorted | Full screen

time	source	agent	rule.description
Apr 15, 2023 @ 20:22:36.388	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:22:31.782	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:20:34.551	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:19:33.193	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:18:26.311	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:17:25.209	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:16:22.304	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:15:40.578	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:14:40.471	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:13:39.361	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:12:38.251	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:11:37.141	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:10:36.031	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:09:34.921	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:08:33.811	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:07:32.701	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:06:31.591	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:05:30.481	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:04:29.371	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:03:28.261	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:02:27.151	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:01:26.041	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 20:00:24.931	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:59:23.821	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:58:22.711	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:57:21.601	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:56:20.491	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:55:19.381	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:54:18.271	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:53:17.161	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:52:16.051	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:51:14.941	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:50:13.831	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:49:12.721	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:48:11.611	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:47:10.501	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:46:09.391	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:45:08.281	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:44:07.171	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:43:06.061	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:42:04.951	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:41:03.841	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:40:02.731	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:39:01.621	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:38:00.511	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:36:59.401	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:35:58.291	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:34:57.181	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:33:56.071	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:32:54.961	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:31:53.851	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:30:52.741	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:29:51.631	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:28:50.521	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:27:49.411	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:26:48.301	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:25:47.191	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:24:46.081	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:23:44.971	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:22:43.861	Urbano-HQ		Successful login to RDP
Apr 15, 2023 @ 19:21:42.751	Urbano-HQ		Successful login to RDP

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5. Custom Detection Engineering

5.1 Custom XML Decoder & pfSense Rule (Rule ID 100101)

A custom XML decoder was engineered to parse pfSense syslog payloads without conflicting with native Wazuh rule signatures. A corresponding custom detection rule (ID 100101 — 'pfSense: Fresh Start Alert') was authored at rule level 5 and mapped to PCI DSS 10.2.4, ensuring perimeter reboot and configuration change events were ingested and searchable in Wazuh's Discover view.

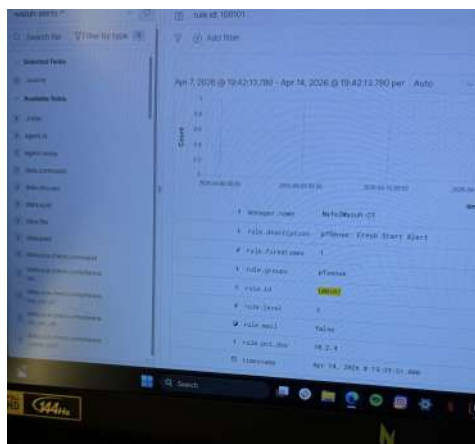


Fig 7 — Wazuh Discover: Custom rule 100101 ('pfSense: Fresh Start Alert') firing from manager Nafe3Wazuh-CT, tagged to pfSense group and PCI DSS 10.2.4.

5.2 File Integrity Monitoring (FIM) — MITRE ATT&CK Mapped

Wazuh FIM was configured to monitor the Kali endpoint's filesystem. A simulated file modification on the desktop triggered rule 550 at level 7, mapped to MITRE ATT&CK T1565.001 (Stored Data Manipulation). The alert captured cryptographic hash changes (MD5, SHA1, SHA256), inode deltas, and the diff of the modification — demonstrating granular, zero-loss file change visibility with compliance tagging (HIPAA 164.312, PCI DSS 11.5, NIST SI.7).

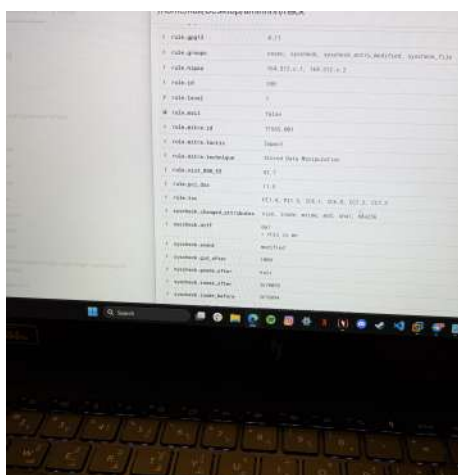


Fig 8 — Wazuh FIM alert detail: Rule 550, MITRE T1565.001 (Stored Data Manipulation), capturing hash changes and file diff on /home/kali/Desktop/aminnnn/hack.

6. Adversary Emulation — Sliver C2 Framework

To validate the full detection stack, an adversary emulation exercise was executed using the Sliver C2 framework from the Kali attacker node. A payload named COMPLEX_SENATOR was delivered to the Windows 10 target (10.10.20.133) via a Python HTTP server. Two C2 sessions were established (one ALIVE, one DEAD — confirming beacon lifecycle tracking), and the 'whoami' command was executed remotely. The running process list on the target included security tools (Windows Defender, Windows SmartScreen) alongside the beacon process, providing a realistic detection challenge.

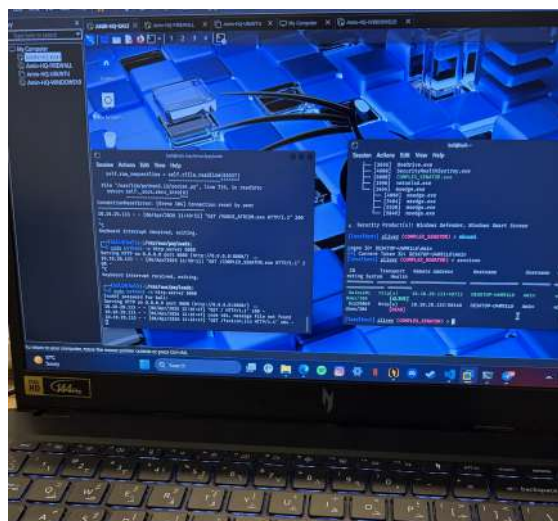


Fig 9 — Sliver C2 (COMPLEX_SENATOR listener): Active session on DESKTOP-U4M91L6 (Windows 10 target), remote whoami executed, process list including security tooling visible.

7. Wazuh SOC Dashboard — Blocked Traffic Visualization

A custom 'Libya SOC' dashboard was built within Wazuh to visualize pfSense-blocked traffic in real time. The dashboard surfaced blocked traffic by source IP (donut chart), event frequency over time (bar chart), and a raw event breakdown table — enabling rapid triage of firewall-level block events correlated to Wazuh rule levels 0–6.

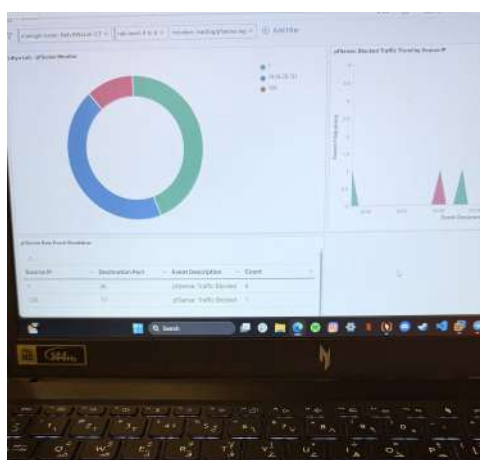


Fig 10 — Libya SOC Wazuh Dashboard: pfSense Blocked Traffic Trend by Source IP with donut breakdown and raw event table filtered to /var/log/pfsense.log.

8. Windows Event Log Analysis

EventLog Analyzer was deployed on the Windows 10 endpoint alongside the Wazuh agent, with two devices registered (DESKTOP-O49I3O6 and DESKTOP-U4M91L6). This provided a secondary log aggregation layer for Windows event channels, complementing Wazuh's EDR telemetry with structured Windows Security log parsing.

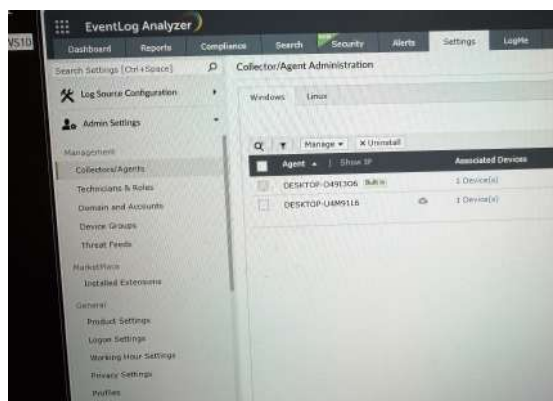


Fig 11 — EventLog Analyzer Collector/Agent Administration: Two Windows endpoints registered for Windows Event Log collection.

9. Key Engineering Challenges Resolved

- **Suricata Checksum Validation:** Virtual NIC checksum offloading caused packet-drop anomalies in Suricata. Resolved by disabling checksum validation in `suricata.yaml`, restoring accurate traffic inspection.
- **rsyslog Pipeline Conflicts:** YAML/JSON serialization errors in the rsyslog configuration prevented Suricata `eve.json` payloads from reaching Wazuh cleanly. Resolved through targeted rsyslog template adjustments and user group permission corrections on the Wazuh agent socket.
- **Custom XML Decoder Collision:** Default Wazuh XML decoders conflicted with pfSense syslog field naming. A custom decoder was engineered to parse pfSense-specific fields without disrupting existing native rule signatures.
- **C2 Payload Delivery (HTTP 404):** Initial Sliver payload delivery via Python HTTP server returned 404 errors due to incorrect working directory. Resolved by aligning the HTTP server root with the payload output path.

10. Outcomes & Demonstrated Competencies

- Deployed a fully functional 4-tier SOC lab: perimeter firewall → IDS/IPS → SIEM → endpoint monitoring
- Engineered custom Wazuh XML decoders and detection rules with compliance mapping (PCI DSS, HIPAA, NIST, MITRE)
- Configured and tuned Suricata IDS/IPS with live alert generation and rsyslog forwarding to Wazuh
- Built a custom Wazuh SOC dashboard with real-time pfSense blocked traffic visualization
- Executed adversary emulation via Sliver C2 — payload delivery, session establishment, remote command execution
- Validated FIM with cryptographic hash tracking mapped to MITRE ATT&CK; T1565.001
- Demonstrated blue-team/red-team duality: built the detection stack, then attacked it to confirm coverage

This project demonstrates end-to-end SOC engineering capability — from raw infrastructure deployment through detection logic authoring to live attack validation. Every component shown was built, broken, debugged, and verified by the author.